



Clinical Practice Guideline

Blood Cultures: Principles of collection

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical and Nursing

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Introduction

Bloodstream Infections (BSIs) cause significant morbidity and mortality. Blood Cultures (BCs) are the current gold standard in the detection of BSIs and are essential for optimising antimicrobial therapy use in patients with sepsis.¹

The aim of this guideline is to assist decision making for when BC collection is indicated and principles to ensure quality BC specimens are obtained.

Risk Statement

Non-compliance with this guideline may:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☐
Other:	☐		

Definitions

Blood culture (BC)	Collection and inoculation of blood into culture medium with the aim of growing pathogenic bacteria or fungi for diagnostic purposes ²
Blood culture set	One aerobic bottle (blue top bottle) and one anaerobic bottle (purple top bottle)

Blood Stream Infections (BSI)	The presence of bacteria or fungi in the bloodstream ²
Cellulitis	Bacterial infection of skin and subcutaneous tissue
Central Venous Access Device (CVAD)	Is a device inserted into the body through a vein to enable the delivery of fluid and other therapies into the blood. CVADs include Central Venous Catheters (CVC), Peripherally Inserted Central Catheters (PICC), implanted ports, non-tunnelled and tunnelled large bore central venous catheters ³
Fever of Unknown Origin (FUO)	Fever of Unknown Origin is defined as. • Fever higher than 38.3°C on several occasions • Duration of fever for at least three weeks Uncertain diagnosis after one week of study in the hospital⁴
Immunocompromised	Neutropenic secondary to chemotherapy, high dose steroids or immunosuppressive therapy, autoimmune and inflammatory conditions, HIV or organ transplantation ⁵
Peripheral Intravenous Catheter (PIVC)	A device that is designed to be inserted into and remain within a peripheral vein ⁶
Uncomplicated cellulitis	• Limited area of involvement • Minimal pain • No systemic signs of illness (e.g., fever, altered mental status, tachypnoea, tachycardia, hypotension) No risk factors for serious illness (e.g., general debility, immunocompromise)⁷
Uncomplicated urinary tract infection	A UTI is classified as uncomplicated if there are no functional or anatomical anomalies in the urinary tract, no renal functional impairment, and no concomitant disease that would promote the UTI ⁸

Key Messages

When to take blood cultures	Clinical suspicion of sepsis (presumed infection and altered physiology, end organ dysfunction). Special circumstances include patients with neutropenic sepsis or indwelling vascular devices (see section 2.3). Fever alone is not predictive of positive blood cultures. Fever combined with signs of physiological instability or rigors/chills increases the likelihood of positive blood cultures.
How to take blood cultures	A BC set is defined as one aerobic bottle (blue top bottle) and one anaerobic bottle (purple top bottle). Two sets (four bottles) of BCs should be collected in patients with

How many?	suspected BSI. In patients with suspected infective endocarditis, three sets (six bottles) must be collected, ideally from three separate venepunctures. ⁹
From where?	A peripheral vein is the preferred site.¹⁰ If a Peripheral Intravenous Catheter (PIVC) is inserted using a strict aseptic technique, BCs can be obtained immediately after insertion. The second BC set must be obtained via a separate venepuncture from a different site. Label each set with the site.² For suspected Catheter-Related Blood Stream Infections (CRBSI) or an unknown source, one set of BCs (two bottles) should be obtained from the suspected infected line as well as another set (two bottles) via a peripheral vein.² When another source of infection is suspected, obtaining blood cultures from a pre-existing Central Vascular Access Device (CVAD) should be discouraged due to the increased likelihood of a false positive.¹¹
How?	Aseptic technique is crucial including hand hygiene and appropriate skin antisepsis. Vial septum of the bottles should be scrubbed with a 70% alcohol impregnated wipe and allowed to dry. ²
How much?	The optimal volume of blood required for each BC bottle is 8-10ml per bottle. Overfilling the bottle will result in a less sensitive culture. ²
When to REPEAT blood cultures	NB: the yield of BCs when the patient has already been started on intravenous antibiotics is low, but may be considered if: 1. There is patient deterioration suggestive of new infection, or non-response to empiric antimicrobial therapy, especially if > 72 hours has elapsed since the previous set of BCs. 2. Initial positive BCs and/or other investigations suggest an endovascular infection (e.g., endocarditis, vascular graft infection) is present. For information on clearance of bacteraemia see section 2.

1. Rationale

1. To diagnose bloodstream infections.
2. Identify the organism(s) causing the BSI and provide susceptibility results to direct appropriate antimicrobial therapy.

2. Indications

- The need for BCs can be determined by identifying infection risk factors and signs of organ dysfunction as per the [SCGOPHCG Adult Sepsis Pathway](#)

Blood cultures should be collected prior to the commencement of antimicrobials; however, in the setting of a deteriorating patient, therapy should NOT be delayed unnecessarily¹².

Traditionally, the “trigger” for obtaining BCs has been fever; however, an isolated fever itself is poorly predictive of bacteraemia. There is only a small correlation between the degree of fever and the likelihood of a BSI. Hypothermia (< 35.5°C) can also be an indicator of sepsis particularly in older adults. Immunosuppressed patients can have a normal temperature and minimal to no clinical evidence of sepsis and still have a BSI and

should have a low threshold for BC.³

Predictors of positive cultures include:

- Clinical suspicion of infection and altered physiology or end organ dysfunction².
- Absence of antibiotics in the last 72 hours
- Working diagnosis of bacteraemia/endocarditis³
 - Two sets (four bottles) of BCs should be collected in patients with suspected BSI.
 - In patients with suspected infective endocarditis, three sets (six bottles) must be collected ideally from three separate venepunctures.⁹

Indications for repeat blood cultures during the admission include:

- Suspected new infection with risk factors for infection and signs of organ dysfunction as per the [SCGOPHCG Adult Sepsis Pathway](#)

Inpatients on antibiotics:

- There is minimal yield from repeating BCs on patients on intravenous or oral antibiotics, if previously collected BC results are pending¹².
- Patients can acquire BSIs in hospital. Therefore, it is reasonable to repeat BCs if there is a presumed new infection, with clear evidence of deterioration or non-response.

Neutropenic Sepsis

Patients with neutropenia and an initial fever must be assessed and treated with empiric broad spectrum antibiotics as per the SCGOPHCG Neutropenic Sepsis guideline.

Specimens must include a minimum of two sets (four bottles) of BCs.

If a Central Venous Access Device (CVAD) is in-situ – one set (two bottles) should be obtained from the CVAD and another set (two bottles) via venepuncture. If multiple CVAD are in situ, specimens should be taken from each device.

For patients with neutropenic sepsis and ongoing fevers, the yield of BCs following initiation of empiric antibiotics and pending results from BCs is very low. Thus, BCs should not be repeated within the first 72 hours UNLESS the patient deteriorates clinically, then two new sets of BCs (four bottles) would be recommended.

Patients with a suspected Catheter-Related Bloodstream Infection (CRBSI)

For patients without neutropenia, the clinical manifestations of a CRBSI may include fever, inflammation or purulence at the catheter insertion site, or systemic signs of sepsis.

In the setting of suspected CRBSI;

- Two BC sets (four bottles) should be obtained. One BC set (two bottles) should be taken via venepuncture, and another set (two bottles) from the suspected infected catheter².
- Each BC set must be clearly labelled noting the location the blood was obtained from (e.g.: “BC via left arm PICC”, “peripheral BC from right antecubital fossa”).
- Clinical improvement within 24 hours following catheter removal is suggestive of CRBSI¹².

Note: be aware of potential complications of CRBSI including septic thrombophlebitis, infective endocarditis and metastatic infection.

Fever of unknown origin

In addition to careful history and examination, three sets (six bottles) of BCs drawn from different sites over a period of at least several hours without administering antibiotics (unless already given) is suggested.

Documenting clearance of bacteraemia

- *Staphylococcus aureus* bacteraemia: One set (two bottles) should be collected 24-48-hours until clearance of the bacteraemia is documented.
- If the patient has bacteraemia/fungaemia due to endovascular infection (e.g. infective endocarditis, vascular graft infection) or any *Staphylococcus aureus* bacteraemia, repeat BCs are required to document clearance of bacteraemia to inform decisions (e.g. removal of indwelling devices or insertion of new devices).
- For bacteraemia without endovascular infection (and not *Staphylococcus aureus*) repeat blood cultures are not necessary unless directed by ID/Micro or if there is clinical evidence of uncontrolled infection despite appropriate antimicrobial therapy (e.g. ongoing sepsis, embolic phenomena, new secondary sites of infection).

2.1 Circumstances in which blood cultures are NOT indicated

In specific patient groups, obtaining BCs does not change management. Blood cultures should NOT be collected in the following groups of patients;

- Patients being discharged from the Emergency Department
- Mild to moderate community acquired pneumonia (unless the patient is immunocompromised or being admitted to hospital)¹³
- Uncomplicated cellulitis¹⁴
- Uncomplicated urinary tract infection¹⁵

In the above settings, the likelihood of a false positive is higher than the likelihood of a positive BC changing practice.¹⁶

3. How to Take Blood Cultures**Safety information**

- Appropriate Personal Protective Equipment (PPE) must be worn when collecting specimens as required under standard and cytotoxic precautions.
- Observe the five moments for hand hygiene and aseptic technique principles at all times during the BC collection.
- Follow the risk assessment outlined in SCGOPHCG Aseptic Technique Policy refer to SCGOPHCG HealthPoint / Directory / Infection Prevention and Control / Infection Prevention and Management Manual / Aseptic Technique Policy.
- The person requesting the specimen is responsible for completing the request form via Computerised Provider Order Entry (CPOE) or manually if CPOE is unavailable.
- The person collecting the specimen is responsible for labelling the specimen at the bedside using the patient identification labels. If labels unavailable, full name and date of birth or full name and Unique Medical Record Number (UMRN) are minimum requirements to avoid compulsory rejection by the laboratory. Ensure the specimen also includes the time, date and the sampled site (e.g. peripheral, CVC, arterial line).

The patient's identification label should be used whenever available. Ensure the barcode on the bottle is not obscured by the patient label. See image 1.

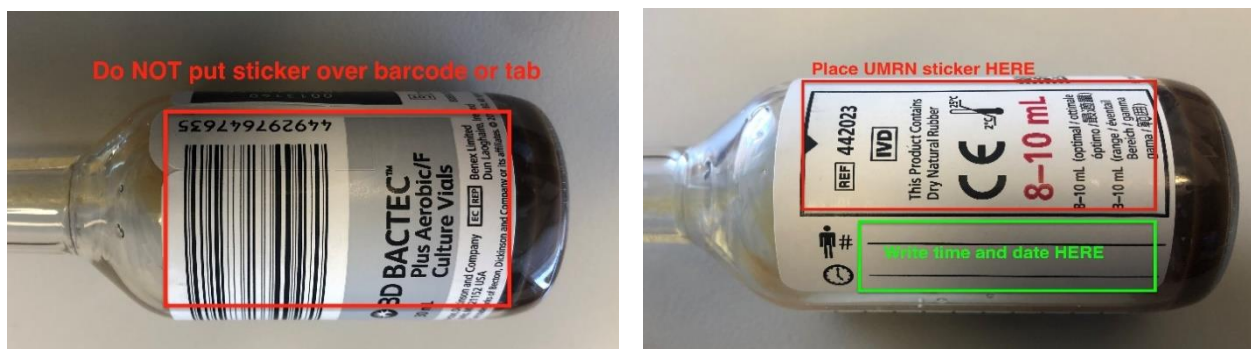


Image 1 How to label a blood culture bottle

Principles to obtaining blood cultures peripherally

- Maintain aseptic technique principles. For a step by step guide on how to obtain BCs, refer to the [SCGOPHCG Specimen Collection for Microbiological Examination Guideline](#).
- If other blood tests are performed, the BCs should be collected before any other specimens.²
- Venepuncture is the preferred option for obtaining BCs as PIVCs are frequently colonized with skin flora, increasing the likelihood of a false-positive BC¹⁷. If BCs are obtained from a PIVC, it is important that the cultures are obtained directly at the time of insertion and an additional set of BCs is obtained from a venepuncture site.¹⁰

Ensure the aerobic bottle (blue top bottle) is collected first, followed by the anaerobic bottle (purple top bottle). See Image 2.



Image 2: One set of blood culture bottles

Ensure 8-10 ml of blood is obtained in each BC bottle.² If insufficient blood (<10mls) is collected, inoculate the entire sample into the aerobic bottle only (blue top bottle)¹⁷.

Principles to obtaining blood cultures via a CVAD

- For a step by step guide on how to obtain BCs from a CVAD, refer to the [SCGOPHCG Specimen Collection for Microbiological Examination Guideline](#).

Taking a BC from a CVAD follows the same principles and preparation as obtaining a BC from a peripheral site. The differences are;

- When blood is drawn through a pre-existing CVAD;
 1. If it is **locked**, discard the first 5 mL of blood
 2. If it is **not locked**, the first 5 mL should be sampled (as the majority of the micro-

organisms in a CRBSI will be adherent on the inside of the line).

- **DO NOT** remove the Positive Pressure Valve (PPV) during blood sampling from a CVAD. See Image 4.



Image 4: Positive Pressure Valve

Related Documents

[SCGOPHCG Clinical Practice Guideline: Non-tunnelled Central Venous Catheters \(CVC\) and Peripherally Inserted Central Catheters \(PICC\)](#)

[SCGOPHCG Infection Prevention and Control Policy Manual: Specimen Collection for Microbiological Examination Guideline](#)

[SCGOPHCG Infection Prevention and Control Policy Manual: Aseptic Technique](#)

[SCGOPHCG Transfusion Policy Manual: Sample Collection and Labelling](#)

[SCGOPHCG Nurse Practice Guideline: Management of Antineoplastic and Hazardous Agent](#)

[SCGOPHCG Sepsis Management CPG](#)

References

- 1) Nannan Panday RS, Wang S, van de Ven PM, Alam N, Nanyakkara PW. Evaluation of blood culture epidemiology and efficiency in a large European teaching hospital. PLoS One. 2019 March; 4(3), doi 10.1471/journal.pone.0214052
- 2) Ntusi N, Lindsey A, Oliver S, Whitelaw A, Mendelson M. Guideline for the optimal use of blood cultures. South African Medical Journal. 2010 December;100(12):839-843
- 3) Ho C, Spry C. Central venous access devices (CVADs) and peripherally inserted central catheters (PICCS) for adult and paediatric patients: a review of clinical effectiveness and safety. In Canadian Agency for Drugs and Technologies in Health, Rapid Response Reports. 2017 April
- 2) Geisler BP, Jilg N, Patton RG, Pietzsch JB. Model to evaluate the impact of hospital-based interventions targeting false-positive blood cultures on economic and clinical outcomes. Journal of Hospital Infection. 2019 March; 102, 438-444. Doi 10.1016/j.jhin.2019.03.012
- 3) Linsenmeyer K, Gupta K, Strymish JM, Dhanani M, Brecher SM, Breu AC. Culture if spikes? Indications and yield of blood cultures in hospitalized medical patients. Journal Hospital Medicine. 2016 May;11(5):336-340.
<https://www.journalofhospitalmedicine.com/jhospmed/article/127949/yield-blood-cultures>
- 4) Bor DH. Approach to the adult with fever of unknown origin. UpToDate. 2020 January
- 5) Fishman J. Approach to the immunocompromised patient with fever and pulmonary infiltrates. UpToDate. 2020 April
- 6) Insertion and Management of Peripheral Intravenous Cannula in Western Australian Healthcare Facilities Policy. MP 0038/16 Version 1.1 Department of Health, 23 August 2017
- 7) Herchline TE. Cellulitis. Medscape. 2019 June. URL:
<https://emedicine.medscape.com/article/214222-overview>
- 8) Wagenlehner FM, Hoyme U, Kaase M, Fünfstück R, Naber KG, Schmiemann G. Uncomplicated urinary tract infections. Dtsch Arztebl International. 2011;108(24):415-423.
doi:10.3238/arztebl.2011.0415

- 9) Lee A, Mirrett S, Reller LR, Weinstein MP. Detection of bloodstream infections in adults: how many blood cultures are needed? *Journal of Clinical Microbiology*. 2007 November; 45(11):3546-3548. doi: 10.1128/JCM.01555-07
- 10) Dawson S. Blood culture contaminants. *Journal of Hospital Infection*. 2014 March; 87:1-10. doi: 10.1016/j.jhin.2014.02.009
- 11) Dreyer A. Blood culture systems: from patient to result. In Azevedo L. Sepsis – An ongoing and significant challenge. Published October 3, 2012, doi.org/10.5772/50139
- 12) Rhodes A, Evans, LE, Alhazzani, W. et al. Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock 2016. *Critical Care Medicine*. 2017 March; 45(3):486-552. doi 10.1097/CCM.0000000000002255
- 13) Metlay JP, Waterer GW, Long AC, Anzueto A, Brozek J, Crothers K, Cooley LA, Dean NC, Fine MJ, Flanders SA, Griffin MR, Metersky ML, Musher DM, Restrepo MI, Whitney CG. Diagnosis and treatment of Adults with Community-acquired pneumonia. An official clinical practice guideline of the American Thoracic Society and Infectious Diseases Society of America. *American Journal of Respiratory and Critical Care Medicine*. 2019 October; 7(1):45-67. doi 10.1164/rccm.201908-1581ST
- 14) Bauer S, Aubert CE, Richli M, Chuard C. Blood cultures in the evaluation of uncomplicated cellulitis. *European Journal of Internal Medicine*. 2016 December; 36:50-56. doi 10.1016/j.ejim.2016.07.029
- 15) Brown JD, Chapman S, Ferguson PE. Blood cultures and bacteraemia in an Australian emergency department: evaluating a predictive rule to guide collection and their clinical impact. *Emergency Medicine Australasia*. 2017 Feb; 29(1):56-62. doi: 10.1111/1742-6723.12696
- 16) Long B, Koyfman A. Best Clinical Practice: Blood culture utility in the Emergency Department. *Journal of Emergency Medicine*. 2016 November; 51(5): 529-539. doi 10.1016/j.jemermed.2016.07.003
- 17) The Joanna Briggs Institute. Recommended Practice. Blood Culture Collection: Adults. The Joanna Briggs Institute EBP Database, JBI@Ovid. 2019; JBI11268

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